OENG1209: Engineering Science

Week 9:

Magnetic

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Today's Topics

- Electron Currents
- Resistance and Ohm's Law
- Magnetic Field
- Magnetic Field of a Current
- Magnetic Forces

Electron Current

The flow of electron

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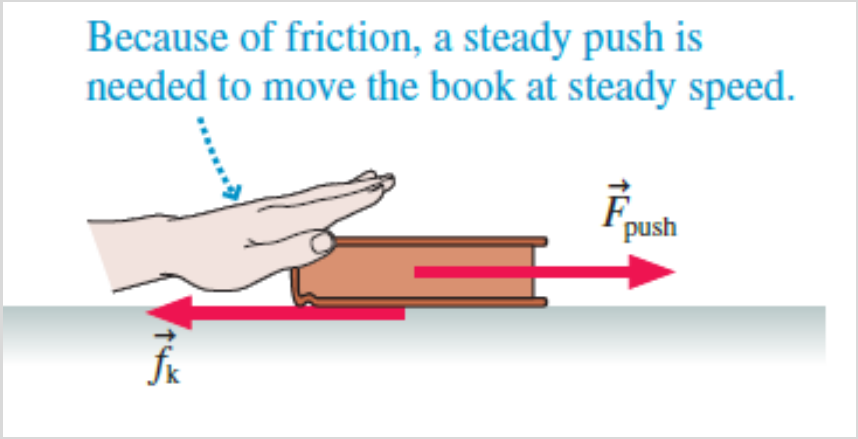
Electron Current



Andre'-Marie Ampere
1775 - 1836

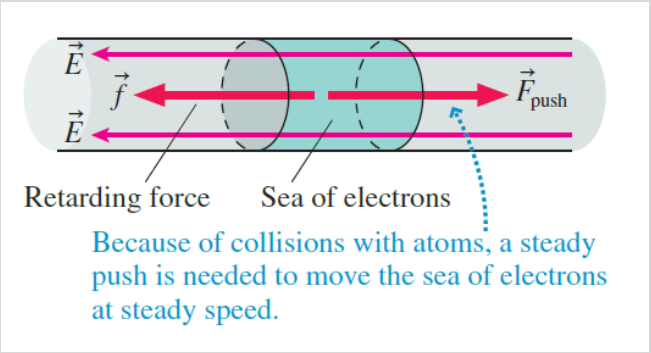
Electron Current

- Consider the action of **sliding a book** across a table to your friend



Electron Current

- The **sea of electrons** is similar to the book. If you **push** the sea of electrons, you create a current of electrons moving through the conductor.
- An **electron current** is a nonequilibrium motion of charges sustained by an internal electric field.



Electron Current

- **Currents** are charges in motion. Hence, it is natural to define current as the rate at which charge moves through a wire.
- If Q is the **total amount of charge** that has moved past a point in the wire, we define the **current I** in the wire to be:

$$I = \frac{dQ}{dt}$$

- For a steady current (which is the focus in this class), the equation above becomes:

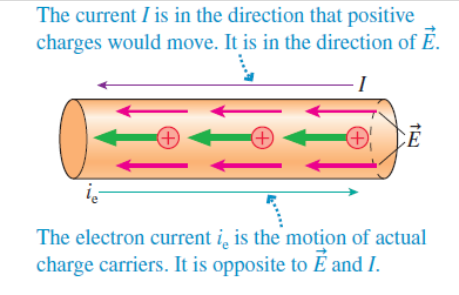
$$I = \frac{Q}{\Delta t}$$

Electron Current

- The unit of current is **Ampere** or Coulomb per second.

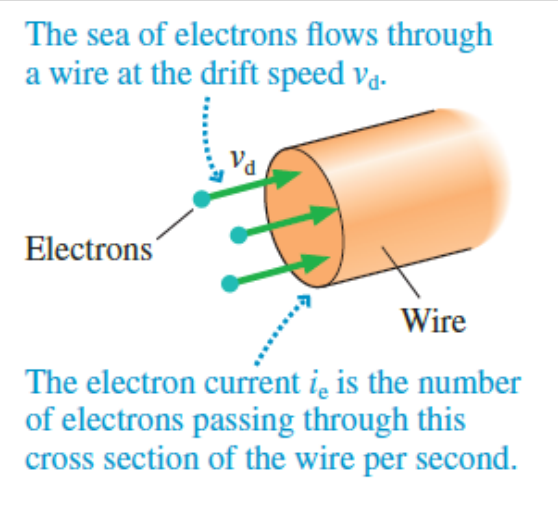
$$1\ ampere = 1\ A = 1\ Coulomb\ per\ second = 1\ C/s$$

- **Household currents** are typically $\approx 1\ A$. Currents in consumer electronics are much less, typically in **milliamps** or **microamps**.
- **The direction of current** is defined to be the direction in which positive charges seem to move. Hence, its direction is the **same as that of the internal electric field $\vec{E}$.**



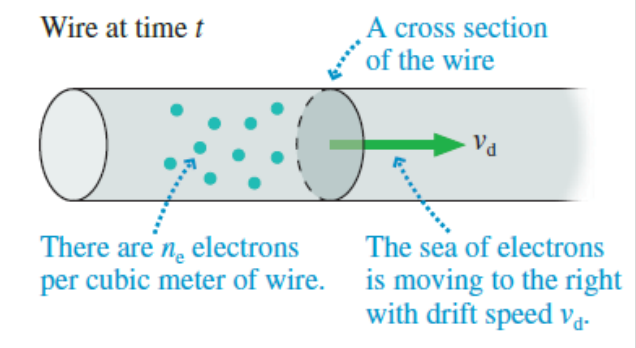
Electron Current

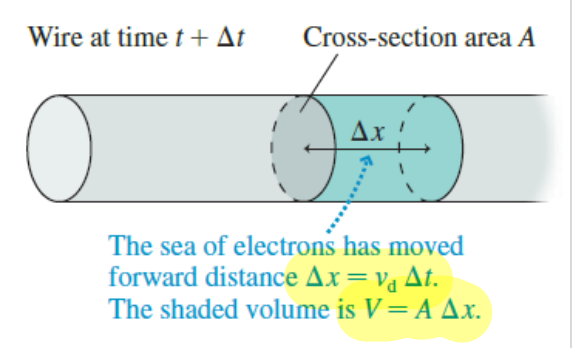
- Another way to examine the electric current is to define it as the number of electrons per second (N_e) that passes through a cross section of a wire (A) or other conductors.



Electron Current

- The speed of the electron current is called the **drift speed v_d**.
- The **drift speed** is typically quite small, typically in the order of magnitude of $10^{-4} m/s$





Electron Current

- Thus, we can express the **electron current i_e (C/s)** as:

$$i_e = n_e A v_d$$

↑ electron/s

n_e is the electron density $(electron/m^3)$

A is the cross sectional area of the wire (m^2).

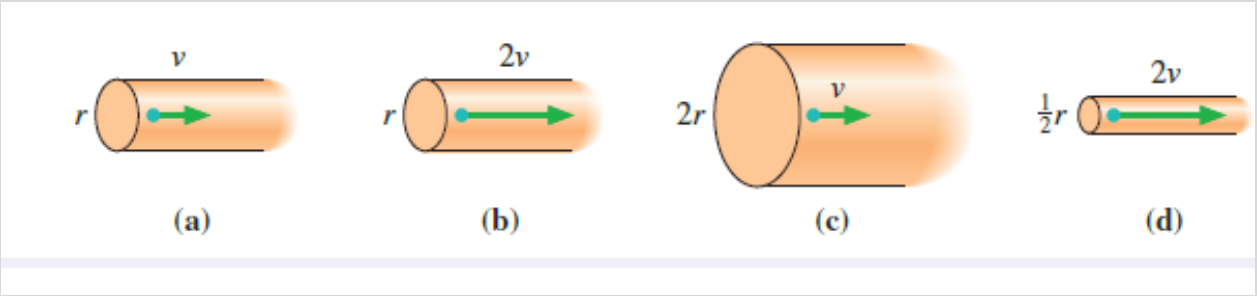
v_d is the drift speed of the electron current (m/s).

Metal	Electron density (m^{-3})
Aluminum	6.0×10^{28}
Copper	8.5×10^{28}
Iron	8.5×10^{28}
Gold	5.9×10^{28}
Silver	5.8×10^{28}

Electron density of some common materials.

Electron Current – Example 1

- **Example:** These four wires are made of the same metal. Rank in order, from largest to smallest, the electron currents i_a to i_d.



Electron Current – Example 2

- **Example:** What is the electron current in a 2.0 mm diameter copper wire if the electron drift speed is $1.0 \times 10^{-4}\ m/s$



Metal	Electron density (m^{-3})
Aluminum	6.0×10^{28}
Copper	8.5×10^{28}
Iron	8.5×10^{28}
Gold	5.9×10^{28}
Silver	5.8×10^{28}

Electron Current

- The two quantities $I\ (A)$ and $i_e\ (electron/s)$ are related by the equation:

$$I = e\, i_e = n_e\, e\, v_d\, A$$

Where e is the charge of a single electron: $1.6 \times 10^{-19}\ Coulomb/electron$

- Thus, we can define a new quantity called the **current density J:**

$$J = \frac{I}{A} = n_e\, e\, v_d\ \ (A/m^2)$$

- Notice that the current density J is also dependent on the drift speed v_d.

Electron Current

- There is another way for us to define the drift speed which is using **Newton's 2nd Law**:

$$F = ma \rightarrow a = \frac{F}{m}$$

$$\rightarrow \boldsymbol{v_d = v_o + \frac{F}{m}\Delta t}$$

- Recall that the force F is the product between **the charge of the electron** $\boldsymbol{e}$ and **the electric field** $\boldsymbol{E}$ such that:

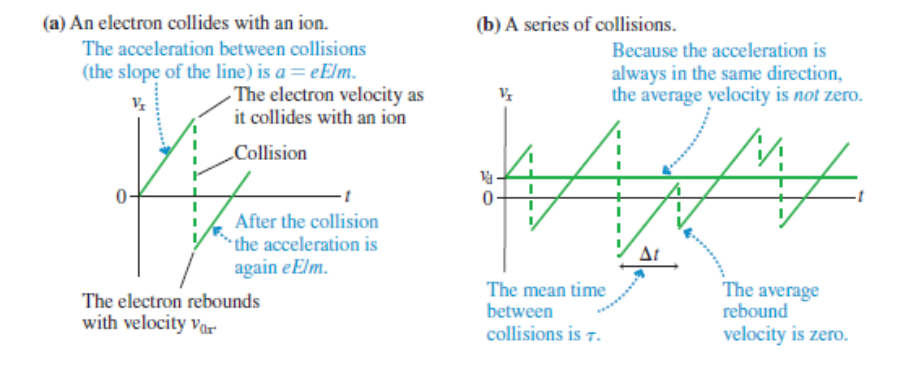
$$F = eE$$

$$\rightarrow \boldsymbol{v_d = v_o + \frac{eE}{m}\Delta t}$$

Electron Current

- Since electrons are constantly colliding with each other, we shall define the time Δt as the average time between 2 consecutive collision.
- Thus, the **average drift speed** of the electron is:

$$\overline{v_d} = \frac{eE}{m}\Delta t$$



Electron Current

- If we use this definition of the average drift speed, the current density is:

$$\boldsymbol{J = n_e e \overline{v_d} = n_e e \left(\frac{eE}{m}\Delta t\right) = \frac{n_e e^2 \Delta t}{m} E}$$

- Here we the can define the conductivity of the material as:

$$\boldsymbol{\sigma = \frac{n_e e^2 \Delta t}{m}}$$

- Thus, the current density can also be written as:

$$\boldsymbol{J = \sigma E}$$

  - Where $\boldsymbol{\sigma}$ is the conductivity specific to the conductor material.
  - $\boldsymbol{E}$ is the electric field strength.

Electron Current

- For many practical applications of current, it will be convenient to use the inverse of the conductivity, called **the resistivity**:

$$\rho = \frac{1}{\sigma} = \frac{m}{n_e e^2 \Delta t}$$

- The **resistivity** of a material tells us how reluctantly the electrons move in response to an electric field.

Material	Resistivity (Ω m)	Conductivity (Ω^{-1}m^{-1})
Aluminum	2.8×10^{-8}	3.5×10^{7}
Copper	1.7×10^{-8}	6.0×10^{7}
Gold	2.4×10^{-8}	4.1×10^{7}
Iron	9.7×10^{-8}	1.0×10^{7}
Silver	1.6×10^{-8}	6.2×10^{7}
Tungsten	5.6×10^{-8}	1.8×10^{7}
Nichrome*	1.5×10^{-6}	6.7×10^{5}
Carbon	3.5×10^{-5}	2.9×10^{4}

Resistivity and conductivity of conducting materials

Electron Current – Example

- **Example:** A 1.0 Ampere current passes through a 1.0-mm diameter aluminum wire. What are the current density and the drift speed of the electrons in the wire? Assume the Electron Density $n_e = 6.0 \times 10^{28}\ electron/m^3$ for aluminum.

Resistance and Ohm's Law

Linking Electric Potential and Current

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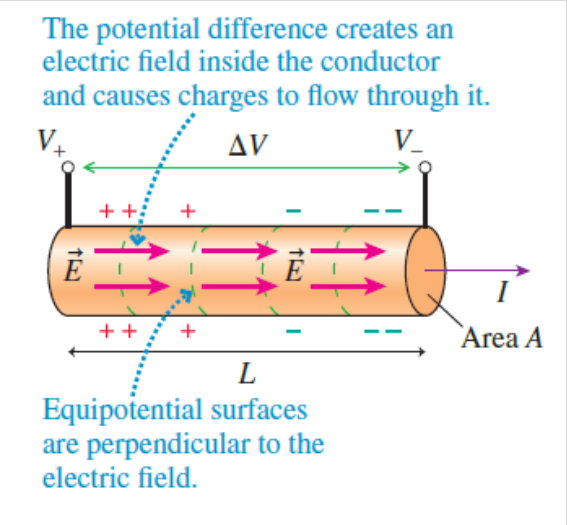
Resistance and Ohm's Law



Georg Simon Ohm
1789 - 1854

Resistance and Ohm's Law

- Next, let us examine a section of wire of length L with an electrical difference $\Delta V = V_+ - V_-$ between the ends.



Resistance and Ohm's Law

- Recall that we previously established the relationship between **the Electric Potential V** and the **Electric field E** as:

$$E_s = -\frac{dV}{ds}$$

- If we only look at **the field strength $|E_s|$** which is constant inside a constant-diameter conductor, the equation above become:

$$\boldsymbol{E = \frac{\Delta V}{\Delta s} = \frac{\Delta V}{L}}$$

- This means the electric field strength inside a constant diameter conductor is simply the potential difference between the ends of the conductor divided by its length.

Resistance and Ohm's Law

- Next, we invoke the relationship between **the current** and **the current density**:

$$I = JA = A\sigma E \rightarrow E = \frac{I}{A\sigma}$$

- Hence we can substitute the field strength as a function of the Electric Potential and Length to obtain:

$$\frac{I}{A\sigma} = \frac{\Delta V}{L} \rightarrow \frac{L}{A\sigma} I = \Delta \mathrm{V}$$

- Here we can define **the electrical resistance** of a conductor R to be:

$$R = \frac{L}{A\sigma} = \frac{L\rho}{A}$$

Resistance and Ohm's Law

- Thus, we have derived **Ohm's Law** which states:

$$\boldsymbol{I = \frac{\Delta V}{R}}$$

  - $\boldsymbol{R}$ is the electrical resistance of the conductor.

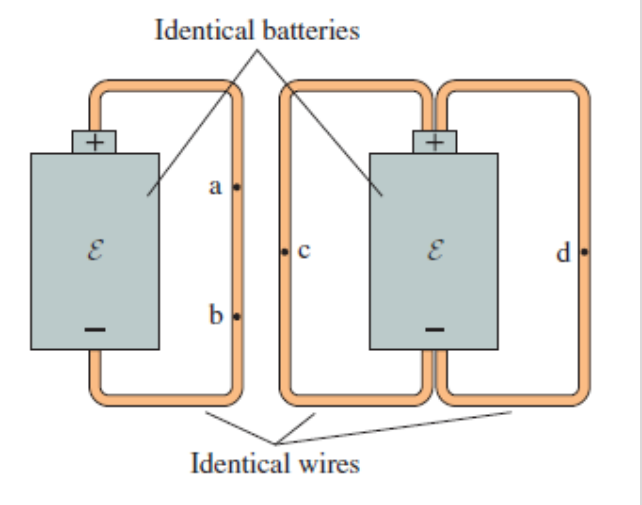
- The unit of electrical resistance is $\boldsymbol{\underline{Ohm}}$.

$$\boldsymbol{1\ ohm = 1\Omega = 1\ V/A}$$

- Note that the equation for R is $R = \frac{L}{A\sigma}$, this means R is directly proportional to the length L and inversely proportional to the cross-sectional area A and the conductivity σ.

Ohm's Law – Example 1

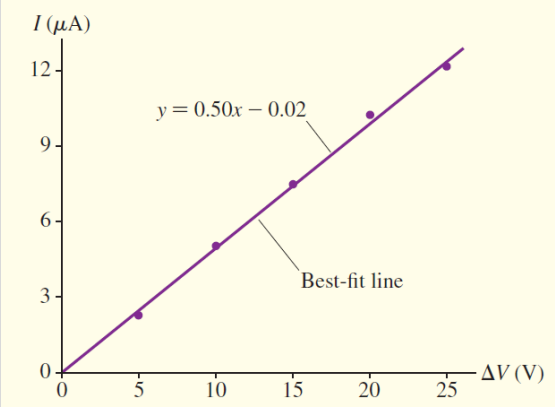
- **Example:** A wire connects the positive and negative terminals of a battery A. Two identical wires connect the positive and negative terminals of battery B (which is identical to battery A). Rank in order from largest to smallest, the currents I_a to I_d at points a to d:



Ohm's Law – Example 2

- **Example:** To determine the conductivity of a leaf, the current is measured when a voltage is applied between two electrodes placed 20 cm apart on a leaf that is 2.5 cm wide and 0.20 mm thick. What is the conductivity of the leaf tissue? Assume the following data were collected using several different voltages:

Voltage (V)	Current (μA)
5.0	2.3
10.0	5.1
15.0	7.5
20.0	10.3
25.0	12.2



Magnetic Field

Linking Electric Potential and Current

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Magnetic Field



MRI Scanning

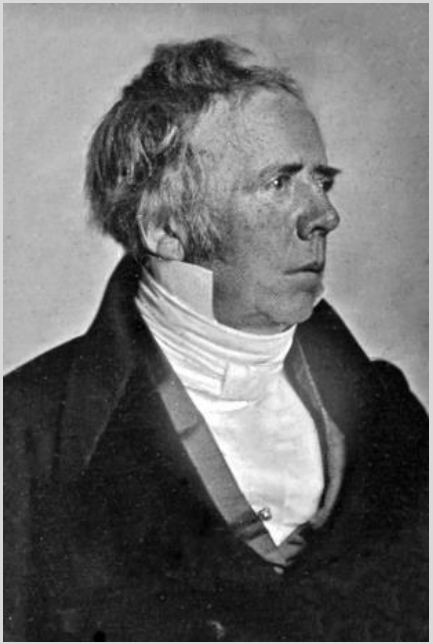


Bullet Train



Compass

Magnetic Field - Discovery

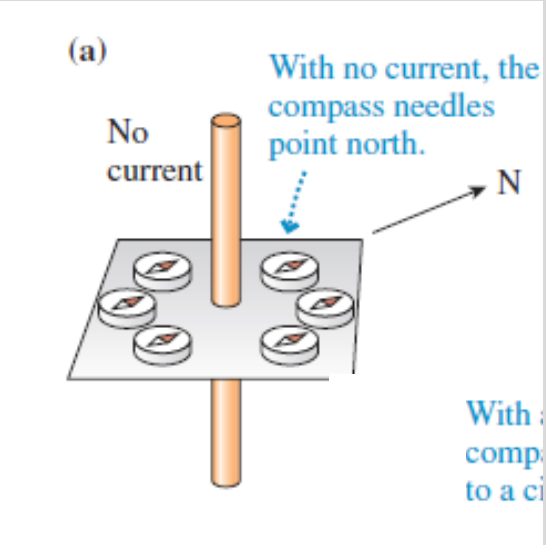


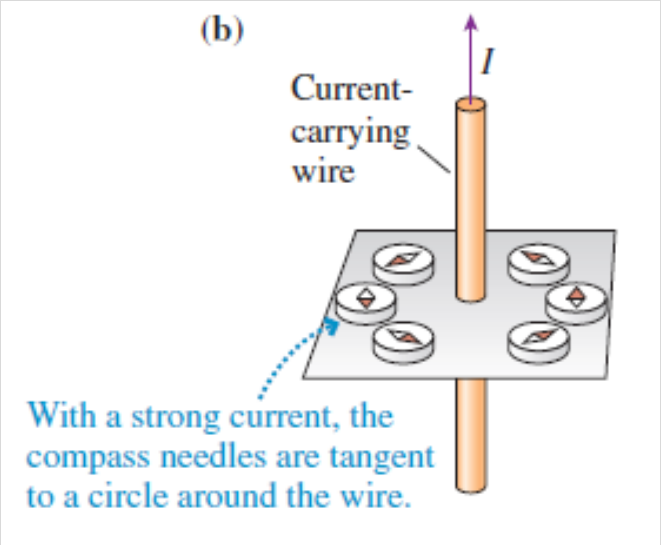
Hans Christian Oersted
1777 - 1851

Magnetic Field - Discovery

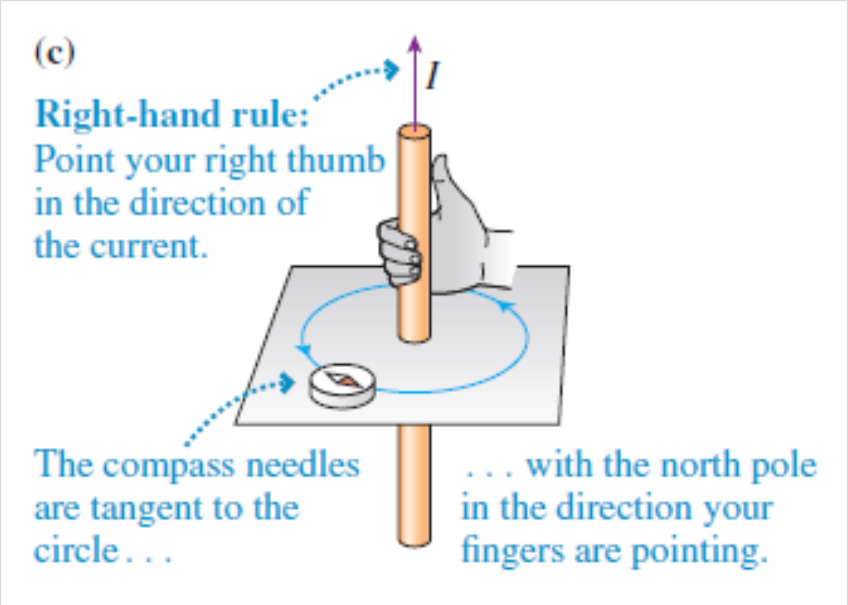
- In 1819, Danish Scientist Hans Christian Oersted discovered the **Magnetic field** during a classroom lecture demonstration.
- In his experiment, **a large current** was produced in a wire. By chance, **a compass** was sitting next to the wire and Oersted noticed that the current caused the compass needle to turn.
- This means magnetism is caused by an electric current.

Magnetic Field - Discovery





Magnetic Field - Discovery

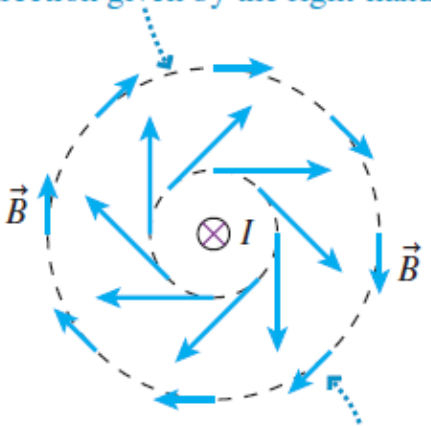


Magnetic Field - Definition

- Previously we introduced the idea of **a $\boldsymbol{field}$** as a way to understand **the long-range electric force**.
- Let us define **the magnetic field $\vec{B}$** as having the following properties:
  - A magnetic field is created **at all points** in the space surrounding a current-carrying wire.
  - The magnetic field at each point is **a vector** with magnitude and direction.
  - The magnetic field **exerts forces** on magnetic poles. The force on a **north pole is parallel to $\vec{B}$** and the force on **a south pole is opposite $\vec{B}$**.

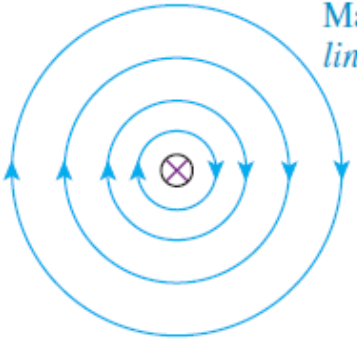
Magnetic Field - Definition

(a) The magnetic field *vectors* are tangent to circles around the wire, pointing in the direction given by the right-hand rule.



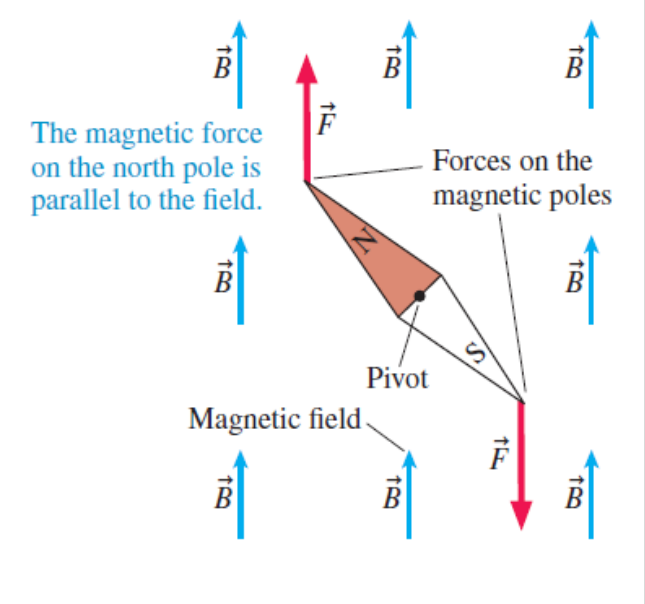
The field is weaker farther from the wire.

(b) Magnetic field *lines* are circles.



Magnetic Field - Definition

- The magnetic field exerts forces on the poles of a compass.



Magnetic Field – Moving Charges

- Since current in a wire generates a magnetic field, and a current is a collection of moving charges, hence, we can guess that moving charges are the source of the magnetic field.
- The **magnetic field** of moving charge is found to be:

$$\vec{B}_{point\ charge} = \left(\frac{\mu_0}{4\pi} \frac{qv\sin(\theta)}{r^2}, \textit{in the direction given by the right hand rule} \right)$$

- The magnetic field strength is measured in **Tesla (T)**.

$$1\ Tesla = 1\ T = 1\frac{N}{A}m$$

  - Typical Refrigerator magnet is about $5.0 \times 10^{-3}\ T$

Magnetic Field – Moving Charges

- This equation is also called the **Biot-Savart law** for a point charge. It is analogous to Coulomb's law.

$$\vec{B}_{point\ charge} = \left(\frac{\mu_0}{4\pi} \frac{qv\sin(\theta)}{r^2}, \textit{in the direction given by the right hand rule} \right)$$

  - μ_0 is the permeability constant of vacuum $4\pi \times 10^{-7}\ T\ m/A$
  - q is the charge of the point charge (Coulomb)
  - v is the velocity of the charged particle (m/s)
  - r is the distance from the point charge to the point at which we want to find $\vec{B}$ (meters)
  - θ is the angle between the $\vec{r}$ and $\vec{v}$ vectors.

Magnetic Field – Moving Charges

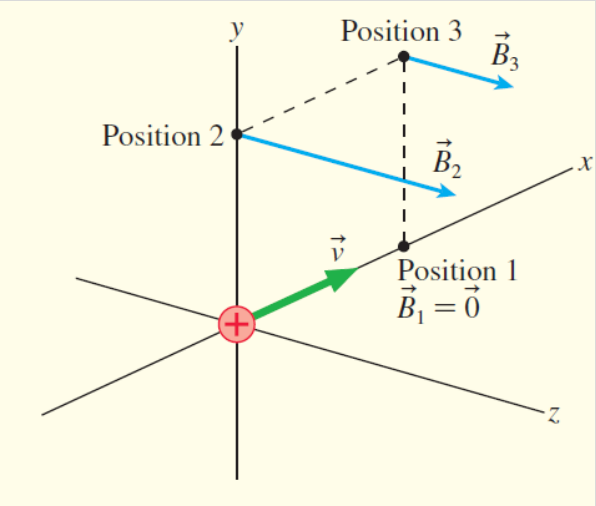
- Note that the **Biot – Savart Law** requires that a charge **be moving** to generate a magnetic field.
- This helps to emphasize a fundamental **distinction** between electric and magnetic fields: **All charges create electric field**, but **only moving charges create magnetic fields**.
- If there are n moving point charges, the net magnetic field is given by the vector sum:

$$\vec{B}_{total} = \vec{B}_1 + \vec{B}_2 + \cdots + \vec{B}_n$$

- This is, again, the principle of superposition.

Magnetic Field – Example 1

- **Example:** A proton moves with velocity $\vec{v} = 1.0 \times 10^7\ \hat{\imath}$ m/s. As it passes the origin, what is the magnetic field at the positions A (1, 0, 0)mm, B(0, 1, 0) mm, C(1, 1, 0) mm?



Magnetic Field – Cross Product

- Similar to Electric Fields, **Magnetic Field is a vector field** which points in the direction given by the right-hand rule.
- Here, we shall use the **Vector Cross-Product** to obtain the full definition of the Biot-Savart law.
- The definition of a Cross Product between 2 vectors $\vec{u}[\mathrm{u_x}, \mathrm{u_y}, \mathrm{u_z}]$ and $\vec{v}[v_x, v_y, v_z]$ is as follows:

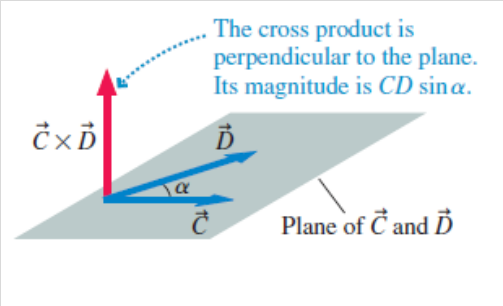
$$\vec{\boldsymbol{u}} \times \vec{\boldsymbol{v}} = [\boldsymbol{u_x}, \boldsymbol{u_y}, \boldsymbol{u_z}] \times [\boldsymbol{v_x}, \boldsymbol{v_y}, \boldsymbol{v_z}]$$

$$= [(\boldsymbol{u_y} * \boldsymbol{v_z} - \boldsymbol{u_z} * \boldsymbol{v_y}), (\boldsymbol{u_z} * \boldsymbol{v_x} - \boldsymbol{u_x} * \boldsymbol{v_z}), (\boldsymbol{u_x} * \boldsymbol{v_y} - \boldsymbol{u_y} * \boldsymbol{u_x})]$$

Magnetic Field – Cross Product

- The definition above is difficult to memorize. Use the following form to achieve the same result.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{m} & \vec{n} & \vec{p} \\ x_a & y_a & z_a \\ x_b & y_b & z_b \end{vmatrix} = \begin{vmatrix} y_a & z_a \\ y_b & z_b \end{vmatrix} \vec{m} - \begin{vmatrix} x_a & z_a \\ x_b & z_b \end{vmatrix} \vec{n} + \begin{vmatrix} x_a & y_a \\ x_b & y_b \end{vmatrix} \vec{p}$$



Magnetic Field – Cross Product

- By definition, the magnitude of the Cross Product between 2 vectors is given by:

$$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}|\sin(\theta)$$

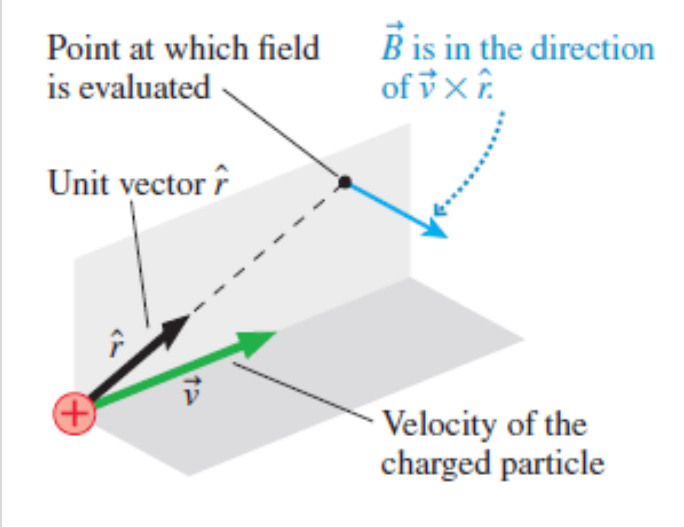
  - Where θ is the angle between the 2 vectors

- Let us define the **unit vector** $\hat{r}$ which points **from charge** q to the point at which we want to evaluate the field. Thus, we have the **complete version of Biot-Savart law**:

$$\vec{B}_{point\ charge} = \left(\frac{\mu_0}{4\pi}\frac{q\vec{v} \times \hat{r}}{r^2}\right)$$

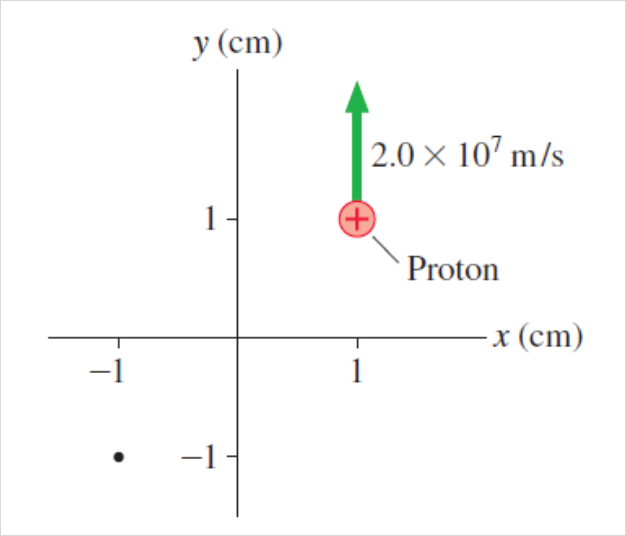
Magnetic Field – Cross Product

$$\vec{B}_{point\ charge} = \left(\frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}\right)$$



Magnetic Field – Example 2

- **Example:** What is the magnetic field at the position of the dot (-1, -1) in the figure below? Give your answer as a vector.



Magnetic Field of a Current

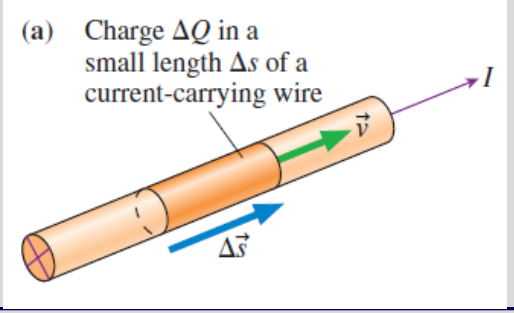
Magnetic field by infinite charges

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Magnetic Field of a Current

- To quantify the **Magnetic Field of a Current**, we first need to write Biot – Savart Law in terms of a **current**.
- Suppose the small amount of moving charge $\boldsymbol{\Delta Q}$ spans the small length Δs. The charge has velocity $\vec{v} = \frac{\Delta \vec{s}}{\Delta t}$.

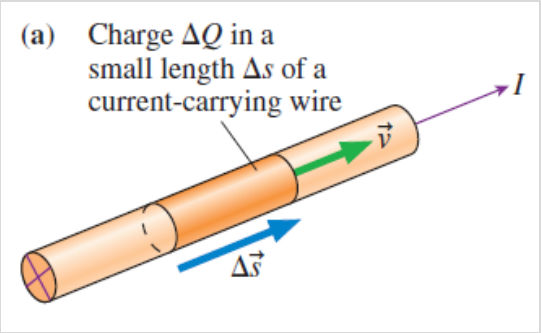
$$(\boldsymbol{\Delta Q})\vec{\boldsymbol{v}} = \boldsymbol{\Delta Q}\frac{\boldsymbol{\Delta \vec{s}}}{\boldsymbol{\Delta t}} = \frac{\boldsymbol{\Delta Q}}{\boldsymbol{\Delta t}}\,\boldsymbol{\Delta \vec{s}} = \boldsymbol{I}\,\boldsymbol{\Delta \vec{s}}$$

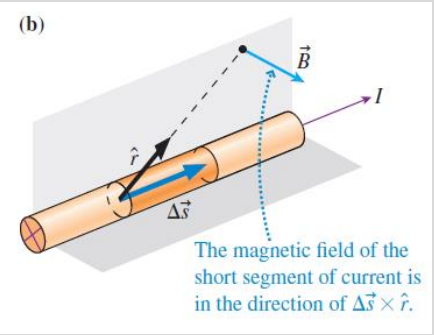


Magnetic Field of a Current

- If we replace $q\,\vec{v}$ in the Biot – Savart law with $I\Delta\vec{s}$, we find that the magnetic field of a very short segment of wire carrying current I is:

$$\vec{B}_{current\ segment} = \frac{\mu_0}{4\pi}\frac{I\,\Delta\vec{s}\times\hat{r}}{r^2}$$



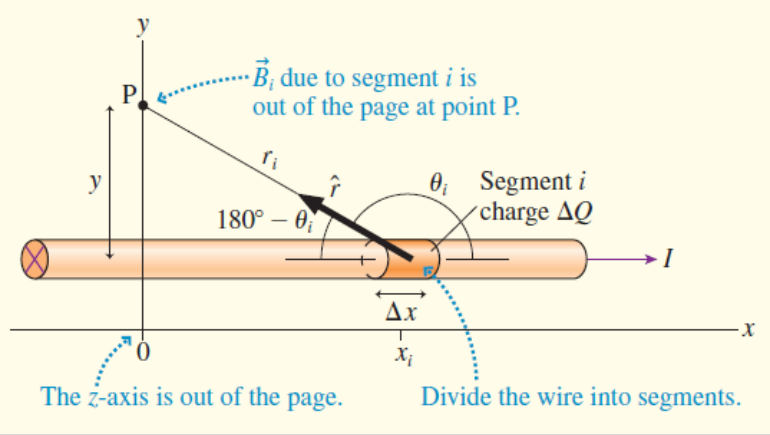


Magnetic Field of a Current

- Next, we align the wire such that it coincides with the positive $x-axis$.

  We can then find the magnetic field <u>at distance r</u> from the wire as:

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,\Delta x\,\sin(\theta_i)}{r_i^2} = \frac{\mu_0}{4\pi}\frac{I\ \sin(\theta_i)}{(x_i^2+y^2)}\Delta x$$



Magnetic Field of a Current

- Using trigonometry, we can express $\boldsymbol{\sin(\theta_i)}$ in terms of $\boldsymbol{x_i}$ *and* $\boldsymbol{y}$ as:

$$\sin(\theta_i) = \frac{y}{r_i} = \frac{y}{\sqrt{x_i^2 + y^2}}$$

- Thus, our Biot – Savart Law for a wire becomes:

$$\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0}}{\boldsymbol{4\pi}} \frac{\boldsymbol{I\, y}}{\left(\boldsymbol{x_i^2 + y^2}\right)^{\frac{3}{2}}} \boldsymbol{\Delta x}$$

Magnetic Field of a Current

- Note that the equation above is only applicable to a small segment of the wire. If we want to have the magnetic field of the whole wire, we should find the sum:

$$B_{wire} = \frac{\mu_0 I y}{4\pi} \sum_i \frac{\Delta x}{\left(x_i^2 + y^2\right)^{\frac{3}{2}}} = \boldsymbol{\frac{\mu_0 I y}{4\pi} \int_{-\infty}^{\infty} \frac{dx}{\left(x_i^2 + y^2\right)^{\frac{3}{2}}}}$$

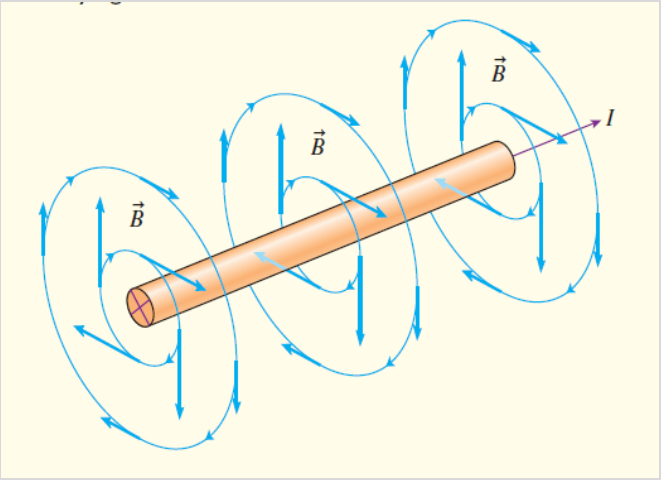
- Solving the Integration above we will obtain:

$$\boldsymbol{B_{wire}} = \frac{\mu_0 I y}{4\pi} \frac{x}{y^2 (x^2 + y^2)^{\frac{1}{2}}} \Big|_{-\infty}^{\infty} = \boldsymbol{\frac{\mu_0 I}{2\pi y}}$$

Magnetic Field of a Current

- Hence, we can conclude that the magnetic field for a wire is:

$$\overrightarrow{B_{wire}} = \left(\frac{\mu_0 I}{2\pi r}, \textit{tangent to a circle around the wire} \right)$$

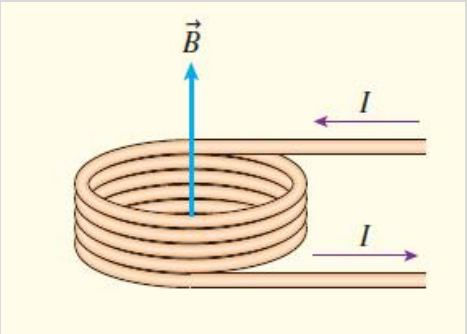


Magnetic Field of a Current

- If the wire is bend into a loop, the equation above becomes (proof available upon request):

$$\vec{B}_{Coil\ Center} = \frac{\mu_0}{2}\frac{NI}{R}$$

  - R is the radius of the loop.
  - N is the number of turns in the loop.

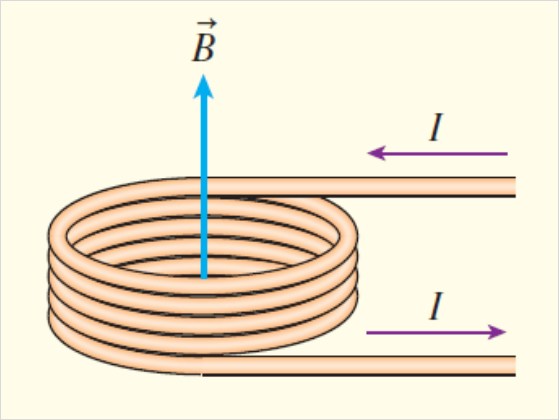


Magnetic Field of a Current – Example 1

- **Example:** A 1.0 meter long, 1.0 mm diameter nichrome heater wire is connected to a 12 V battery. What is the magnetic field strength 1.0 cm away from the wire? Assume the resistivity of nichrome is: $\rho = 1.50 \times 10^{-6}\ \Omega m$.

Magnetic Field of a Current – Example 2

- **Example:** What current is needed in a 5-turn, 10-cm diameter coil to cancel the Earth's magnetic field at the center of the coil? Given that the Earth's magnetic field is about $5 \times 10^{-5}\ T$.



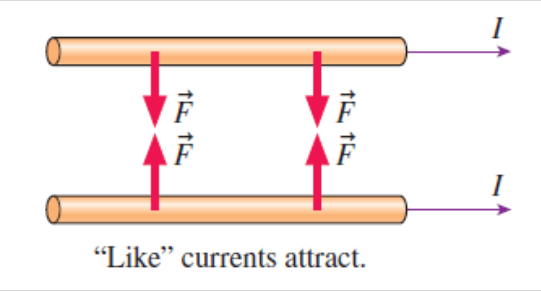
Magnetic Force

On a moving charge and on a current carrying wire

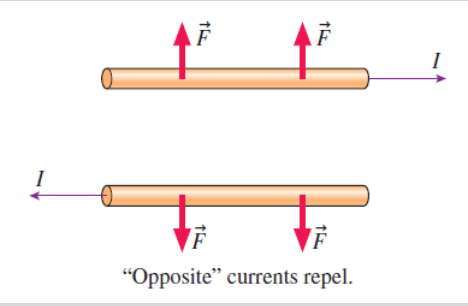
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Magnetic Force on a Moving Charge

- **Oersted** discovered that a current passing through a wire causes a **magnetic torque** to be exerted on a nearby compass needle.
- **Ampere** reasoned that current was acting like a magnet and conducted an experiment.



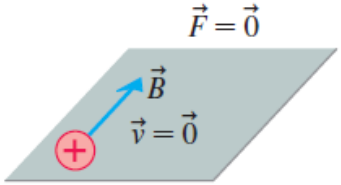
"Like" currents attract.



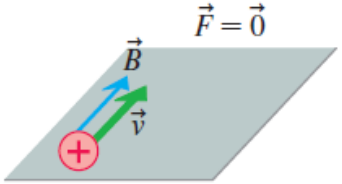
"Opposite" currents repel.

Magnetic Force on a Moving Charge

- Ampere's experiment implies that **a magnetic field exerts a force on a moving charge**.
- It turns out magnetic force depends not only on the charge's velocity but also on how the velocity vector is oriented relative to the magnetic field

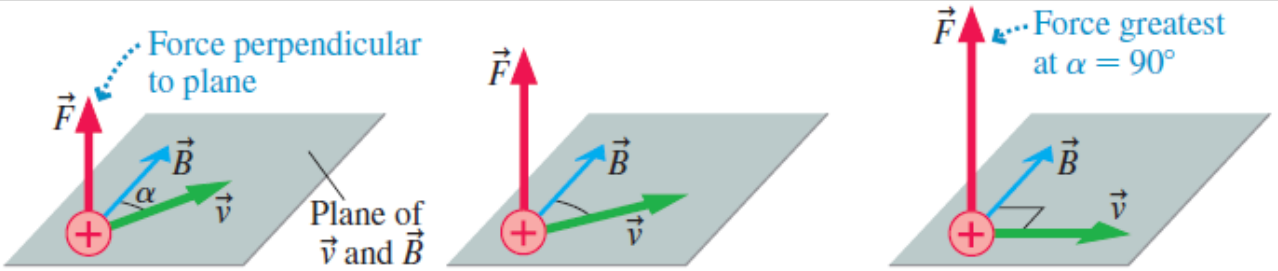


There is no magnetic force on a charged particle at rest.



There is no magnetic force on a charged particle moving *parallel* to a magnetic field.

Magnetic Force on a Moving Charge



As the angle α between the velocity and the magnetic field increases, the magnetic force also increases. The force is greatest when the angle is 90°. The magnetic force is always perpendicular to the plane containing $\vec{v}$ and $\vec{B}$.

Magnetic Force on a Moving Charge

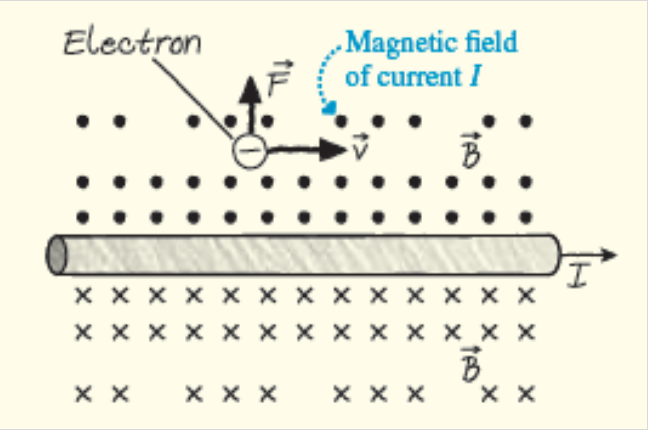
- The **magnetic force** on a charge q as it moves through a **magnetic field** $\vec{B}$ with **velocity** $\vec{v}$ can be written as:

$$\vec{F}_{on\,q} = q\,\vec{v} \times \vec{B} = qvB\,\sin(\alpha)$$

  - q is the charge of the particle (Coulomb)
  - v is the velocity of the particle (m/s)
  - B is the ambient magnetic field (T)
  - α is the angle between the velocity vector and the magnetic field vector.

Magnetic Force on a Moving Charge – Example 1

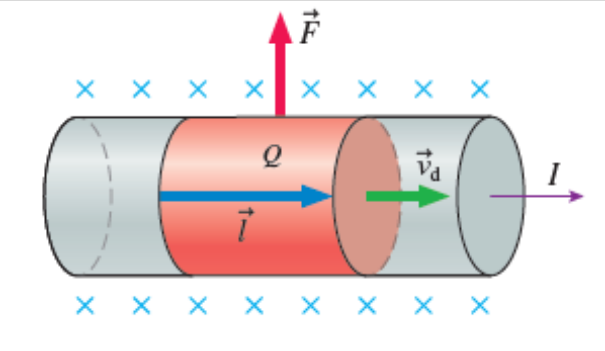
- **Example:** A long wire carries a 10A current from left to right. An electron 1.0 cm above the wire is traveling to the right at a speed of $1.0 \times 10^7 m/s$. What are the magnitude and direction of the magnetic force on the electron?



Magnetic Force on a wire

- A current is moving charge, hence, the magnetic force on a current-carrying wire is simply the **net magnetic force** on all the charge carriers in the wire.
- Because the magnetic force is proportional to q, the net force on all the charge carriers in the wire is the force on the net charge:

$$\vec{F} = Q\,\vec{v_d} \times \vec{B}$$



Magnetic Force on a wire

- By definition, the current I is defined as:

$$I = \frac{Q}{\Delta t} \rightarrow Q = I\,\Delta t$$

- At the same time, the charge carriers have drift speed v_d so they can move a length L within a time Δt which is given by the equation:

$$\Delta t = \frac{\vec{L}}{\overrightarrow{v_d}} \rightarrow \overrightarrow{v_d} = \frac{\vec{L}}{\Delta t}$$

- Substituting the 2 equations above to the Magnetic Force equation, we have:

$$\boldsymbol{\vec{F}_{wire} = I\,\vec{L} \times \vec{B} = I\,L\,B sin(\alpha)}$$

Magnetic Force on a wire - Example

- **Example:** The 0.10 T uniform magnetic field is horizontal, parallel to the floor. A straight segment of 1.0 mm diameter copper wire, also parallel to the floor, is perpendicular to the magnetic field. What current through the wire, and in which direction, will allow the wire to "float" in the magnetic field? Assume the density of copper is $8920\ kg/m^3$.

